



ITAI 2376

A03 Neural Network Zoo

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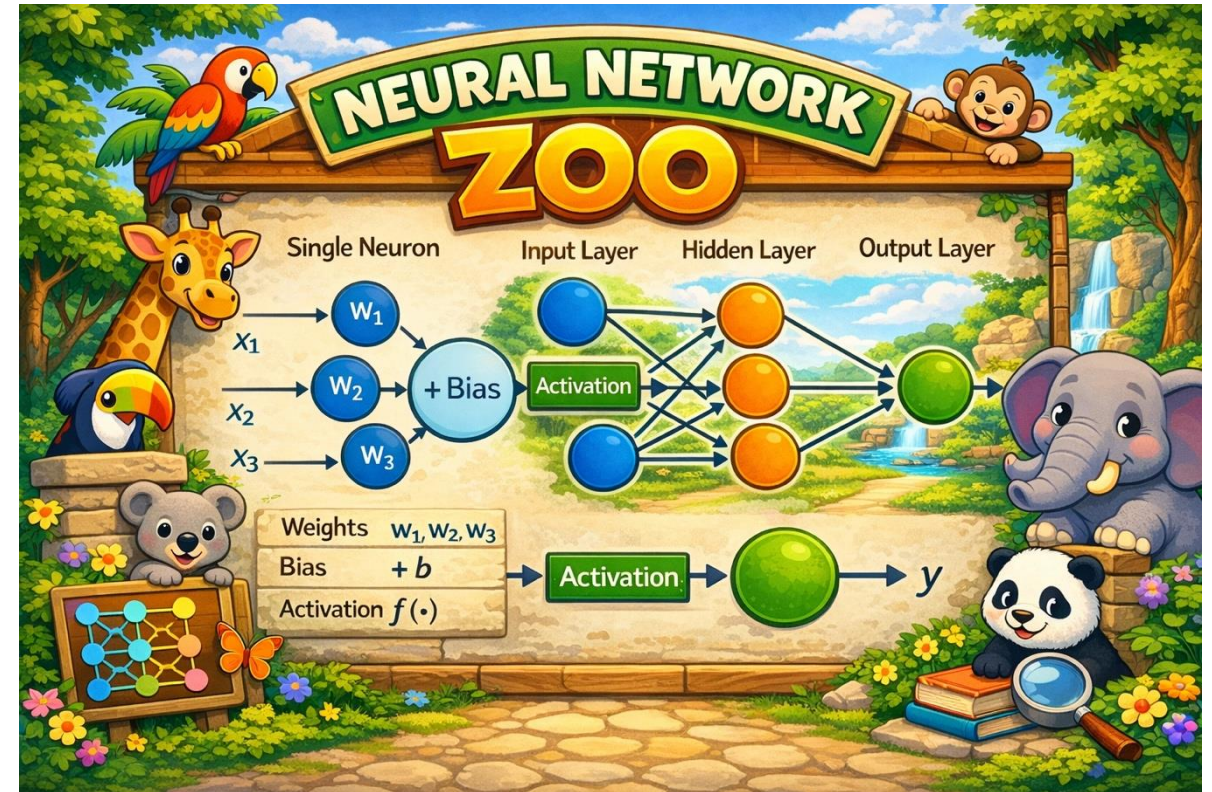
Introduction

Welcome to the Neural Network Zoo. Today, I am going to treat neural networks like animals you can visit, observe, and compare. The goal is not to memorize complex math. The goal is to understand why different networks exist and what each is good at, just as different animals are built for different habitats.

By the end, I will be able to look at a task like recognizing objects in photos or predicting the next word in a sentence and make a smart guess about which network animal belongs there.

Neural Networks and Neurons

- A neural network is a system composed of many small units called neurons, also called nodes. Each neuron receives numerical inputs, assigns different weights to emphasize some inputs over others, adds a small adjustment called a bias, and then decides what to pass forward using an activation step. One neuron is simple, but many neurons together can learn patterns that are hard to describe with rules.
- Neurons are arranged in layers. The input layer receives data, the hidden layers perform most of the learning, and the output layer produces the final prediction. During training, the network adjusts its weights, so the output gets closer to the correct answer across many examples.



The Zoo concept



- In a real zoo, you do not judge a dolphin by how well it climbs a tree. You ask what it was built to do. The Neural Network Zoo works the same way. Each network type becomes an animal with a personality that reflects its strengths.
- This is helpful because it turns confusing names into memorable stories. When you hear "Convolutional neural network," you can picture a cheetah spotting patterns quickly in images. When you hear "Recurrent neural network," you can picture an animal following a trail step by step, remembering what happened a moment ago.

Meet the animals

Zoo map

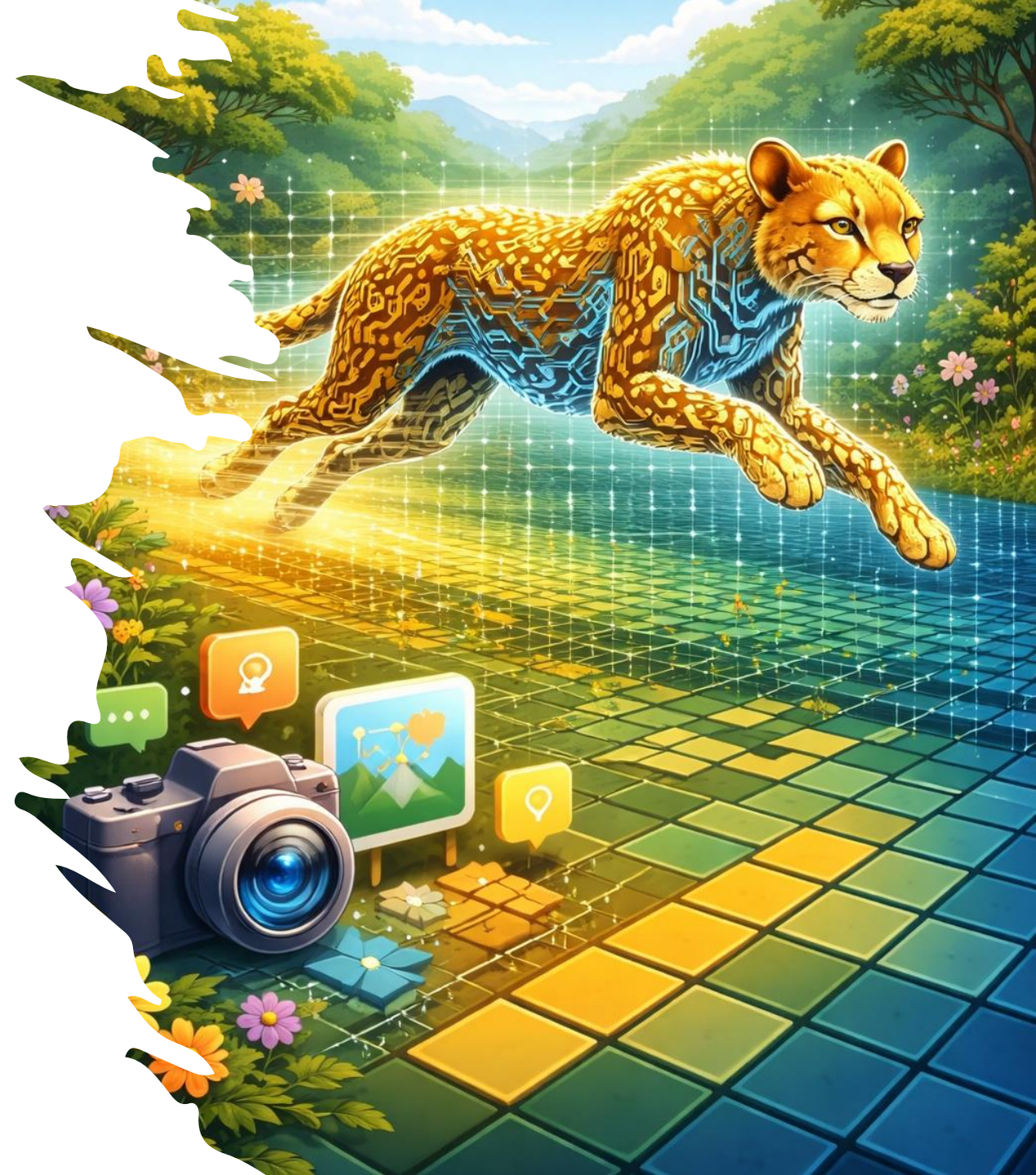
Here is the zoo map, each animal represents a network and its personality matches what it does well.

- **CNN Cheetah**
Fast at scanning images for patterns like edges, shapes, and textures
- **RNN Raccoon**
Good at following sequences like text, audio, or sensor readings
- **LSTM Lemur**
A stronger memory version of the RNN, better when the important clue happened far back
- **Autoencoder Owl**
Learns a compact summary of data, useful for compression and anomaly detection
- **GAN Chameleon**
Creates realistic new samples, like images that look like the training set



CNN Cheetah / The image hunter

- For my own zoo animal, I chose the CNN Cheetah. I picked it because it connects to real moments in daily life, like sorting photos on my phone, spotting a face in a crowd, or identifying objects in a picture. CNNs are known for image classification and object recognition, so the examples feel natural and easy to picture.
- I think of the CNN as a pattern hunter. It examines an image in small parts and learns which details matter. At first, it notices simple shapes like edges and lines, and later it puts those pieces together to recognize larger patterns and whole objects.
- Common places you can see the CNN Cheetah at work
 - Recognizing and labeling what is in a photo
 - Finding and identifying objects within an image, not just naming the whole picture



Conclusion what I learned

- This zoo approach helped me understand neural networks in a way that actually sticks. Instead of feeling like everything was the same, I started to see clear differences in what each network is designed to do. The CNN Cheetah made the most sense to me because spotting visual patterns feels concrete and easy to imagine.
- I also learned that there is no single best network for every task. Each has strengths and limitations, and the best choice depends on the data and the goal. After going through this, I feel more confident explaining the basic idea of a neuron, how layers work together, and why different network types exist in the first place.



References and sources



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